Parameter Flow Matching for Probabilistic Physics Emulation: A Meta-Learning Approach

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Abstract

Efficient and accurate simulation of physical systems is a cornerstone of scientific computing. Classical numerical solvers for partial differential equations (PDEs) offer high precision but incur significant computational costs that scale poorly with resolution and ensemble size. While neural surrogates promise substantial acceleration, most existing architectures are deterministic, collapsing the full predictive distribution of chaotic systems into point estimates. This limitation is fundamentally at odds with the goal of robust uncertainty quantification (UQ), particularly in regimes where long-term deterministic prediction is theoretically bounded. In this work, we propose *Parameter Flow Matching* (PFM), a generative modeling framework that lifts the generative process from the data space of solution fields to the parameter space of a neural regressor. PFM employs a two-stage procedure: first, a meta-learned feature extractor optimizes a basis space wherein the mapping from current to future states is linear; second, a conditional Flow Matching model learns the distribution of the optimal linear weights given the current state. We validate PFM on the chaotic one-dimensional Kuramoto–Sivashinsky (KS) equation, benchmarking it against a deterministic baseline, pixel-space Flow Matching, and Latent Flow Matching. On a challenging prediction task, PFM achieves a mean squared error (MSE) of 0.022, comparable to the deterministic baseline (0.014), while providing calibrated ensemble spread and avoiding the mode collapse and hallucinations observed in data-space diffusion models.

1 Introduction

The simulation of complex physical systems governed by nonlinear partial differential equations (PDEs) is central to disciplines ranging from climate modeling to plasma physics. Traditional numerical methods, such as spectral schemes or finite difference methods, provide rigorous convergence guarantees but are computationally intensive. For chaotic systems, characterizing statistical behavior requires large ensembles of simulations, making high-resolution numerical integration prohibitively expensive.

Scientific machine learning (SciML) has introduced neural surrogates—such as Physics-Informed Neural Networks (PINNs) [Raissi et al., 2019] and Neural Operators [Kovachki et al., 2023]—to approximate the solution operator of PDEs. While these models achieve significant speed-ups, they are predominantly trained with pointwise L_2 objectives, yielding deterministic point predictions. In the context of chaotic dynamics, where small perturbations in initial conditions grow exponentially, a

deterministic surrogate minimizes the mean squared error by averaging over possible futures, often resulting in blurred predictions that fail to capture the underlying physics.

To address this, we decompose the total error of a surrogate model, E_{total}, into three components:

$$E_{\text{total}} \approx E_{\text{aleatoric}} + E_{\text{epistemic}} + E_{\text{misspecification}}. \tag{1}$$

Deterministic models obscure this structure. We argue that for robust forecasting, a surrogate must model the *predictive distribution* rather than a single trajectory. We hypothesize that modeling this uncertainty is more effective in the parameter space of a regressor than in the high-dimensional state space of the PDE solution.

Contributions.

1. We introduce *Parameter Flow Matching* (PFM), a framework that applies conditional Flow Matching to the last-layer weights of a meta-learned feature extractor.
2. We derive a theoretically grounded training objective for the feature extractor based on ridge regression with a Gaussian prior, utilizing trace regularization to control feature energy.
3. We evaluate PFM on the 1D Kuramoto–Sivashinsky (KS) equation, demonstrating that it retains the accuracy of deterministic baselines while providing physically consistent uncertainty quantification, outperforming pixel-space and latent-space generative baselines.

2 Related Work

Neural Surrogates. Operator learning methods, such as the Fourier Neural Operator (FNO) [**?**] and DeepONet [**?**], map infinite-dimensional function spaces. While effective at zero-shot generalization, their standard formulations are deterministic. Probabilistic extensions often rely on Bayesian Neural Networks or ensemble methods, which can be computationally expensive to train or sample.

Generative Modeling in Physics. Generative approaches, including diffusion models and Continuous Normalizing Flows (CNFs), have been applied to super-resolution and weather forecasting (e.g., [Price et al., 2025, Lippe et al., 2023]). These methods typically operate in the "pixel" (grid) space. Flow Matching [**?**] offers a simulation-free alternative to training CNFs by regressing a vector field that transports a base distribution to the data distribution. However, learning flow fields directly in the high-dimensional state space of a PDE is challenging due to the complex topology of the solution manifold.

Meta-Learning and Hypernetworks. Our approach draws inspiration from hypernetworks, where a secondary network generates the weights of a primary network. Unlike standard hypernetworks that regress weights deterministically, PFM treats the weights as random variables conditioned on the system state, learning their distribution via a flow-based generative model.

3 Problem Formulation

We consider the one-dimensional Kuramoto–Sivashinsky (KS) equation, a canonical model for spatiotemporal chaos:

$$\frac{\partial u}{\partial t} + \frac{\partial^4 u}{\partial x^4} + \frac{\partial^2 u}{\partial x^2} + u\frac{\partial u}{\partial x} = 0, \quad x \in [0, L],\, t \in \mathbb{R}^+, \tag{2}$$

subject to periodic boundary conditions $u(x,t) = u(x+L,t)$. The system exhibits rich dynamical behavior, including bifurcations and transitions to chaos, depending on the domain size L.

We address the problem of probabilistic one-step prediction. Let $\mathbf{u}_t \in \mathbb{R}^N$ denote the discretized state vector at time t. We seek to model the conditional distribution

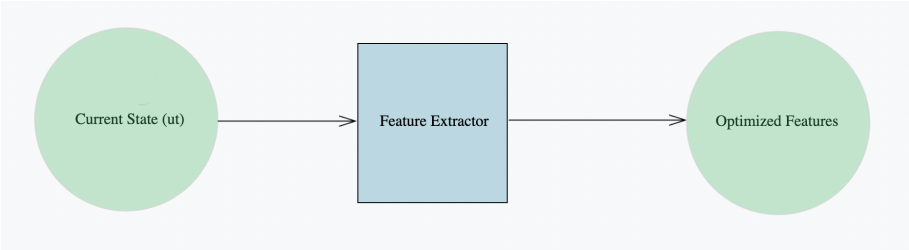
$$p(\mathbf{u}_{t+\Delta t} \mid \mathbf{u}_t), \tag{3}$$

where Δt is a fixed forecast horizon. The goal is to capture the epistemic uncertainty arising from finite data and the inherent instability of the chaotic attractor.

4 Methodology: Parameter Flow Matching

PFM decomposes the forecasting problem into two stages: (1) learning a deterministic basis where the dynamics are locally linearizable, and (2) modeling the distribution of the linear operators (weights) that map the current state to the future state.

4.1 Stage 1: Meta-Learned Feature Extraction



Let $\Phi_\theta : \mathbb{R}^N \to \mathbb{R}^{C\times N}$ be a neural feature extractor parameterized by θ, mapping the input state $\mathbf{u}_t$ to a set of C feature maps. We posit that the transition to $\mathbf{u}_{t+\Delta t}$ can be approximated by a linear transformation of these features. For a given pair $(\mathbf{u}_t, \mathbf{u}_{t+\Delta t})$, let $\mathbf{F} = \Phi_\theta(\mathbf{u}_t) \in \mathbb{R}^{N\times C}$ (reshaped) and $\mathbf{y} = \mathbf{u}_{t+\Delta t} \in \mathbb{R}^N$. We assume a linear relationship:

$$\mathbf{y} \approx \mathbf{F}\mathbf{w}, \tag{4}$$

where $\mathbf{w} \in \mathbb{R}^C$ are the specific weights for this transition. The optimal weights $\mathbf{w}^*$ are obtained via Ridge Regression:

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} \left(\|\mathbf{F}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda\|\mathbf{w}\|_2^2\right) = (\mathbf{F}^\top\mathbf{F} + \lambda\mathbf{I})^{-1}\mathbf{F}^\top\mathbf{y}. \tag{5}$$

To train Φ_θ, we do not simply minimize the reconstruction error. We incorporate a regularization term derived from the Gaussian prior on $\mathbf{w}$, which manifests as a penalty on the feature energy (trace of the covariance). The meta-objective for θ is:

$$\mathcal{L}_{\text{FE}}(\theta) = \mathbb{E}_{(\mathbf{u}_t,\mathbf{u}_{t+\Delta t})}\left[\|\mathbf{F}\mathbf{w}^* - \mathbf{y}\|_2^2 + \sigma^2 \text{Tr}(\mathbf{F}^\top\mathbf{F})\right], \tag{6}$$

where σ^2 relates to the prior variance of the weights. This objective ensures that the learned basis functions are informative yet compact, preventing pathological scaling that would destabilize the subsequent weight generation.

4.2 Stage 2: Conditional Flow Matching in Weight Space

Once Φ_θ is trained and frozen, we compute the optimal weights $\mathbf{w}_i^*$ for every transition pair i in the training set using Eq. (5). This transforms our dataset from pairs of states to pairs of states and optimal weights: $\mathcal{D} = \{(\mathbf{u}^{(i)}, \mathbf{w}^{*(i)})\}$.

We model the conditional distribution $p(\mathbf{w} \mid \mathbf{u})$ using Flow Matching. We define a probability path $p_t(\mathbf{z} \mid \mathbf{u})$ where $\mathbf{z} \in \mathbb{R}^C$ represents the latent weight variable. We choose a Gaussian probability path that interpolates between a standard normal prior $p_0(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ and the target Dirac distribution centered at $\mathbf{w}^*$. The conditional vector field $v_t(\mathbf{z} \mid \mathbf{u})$ is parameterized by a neural network $\mathbf{v}_\phi(\mathbf{z}, t, \mathbf{u})$. The objective is to regress the target velocity field that generates the interpolation $\mathbf{z}_t = (1-t)\mathbf{z}_0 + t\mathbf{w}^*$:

$$\mathcal{L}_{\text{PFM}}(\phi) = \mathbb{E}_{t\sim\mathcal{U}[0,1],\mathbf{z}_0\sim p_0,(\mathbf{u},\mathbf{w}^*)\sim\mathcal{D}}\left[\|\mathbf{v}_\phi(\mathbf{z}_t, t, \mathbf{u}) - (\mathbf{w}^* - \mathbf{z}_0)\|_2^2\right]. \tag{7}$$

At inference time, given a new state $\mathbf{u}$, we sample weights by solving the ODE:

$$\frac{d\mathbf{z}}{dt} = \mathbf{v}_\phi(\mathbf{z}, t, \mathbf{u}), \quad \mathbf{z}(0) \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{8}$$

The final prediction is constructed as $\hat{\mathbf{u}}_{t+\Delta t} = \Phi_\theta(\mathbf{u})\mathbf{z}(1)$.

5 Implementational Details

5.1 Pixel-Space Flow Matching (FM)

The first baseline, Pixel-Space Flow Matching (FM), models the conditional velocity field $v_\phi(\mathbf{x}_t, t, \mathbf{u}_t)$ directly in the state space $\mathbb{R}^N$. The architecture employs a conditional 1D U-Net backbone designed to process the noisy interpolant $\mathbf{x}_t$ while conditioning on the current clean state $\mathbf{u}_t$ and the integration time t.

The U-Net features a multi-scale encoder-decoder structure with four resolution levels. The channel width doubles at each stage according to multipliers $[1, 2, 4, 8]$ relative to a base width of 32 channels. An initial 3×1 convolution projects the scalar input to the base dimension.

- **Encoder:** Each resolution level consists of two residual convolutional blocks followed by a strided 3×1 convolution for downsampling, which halves the spatial resolution while increasing channel depth.
- **Bottleneck:** At the coarsest resolution, the representation is processed by two additional residual blocks to capture global context.
- **Decoder:** The decoder mirrors the encoder path. Features are upsampled via nearest-neighbor interpolation and refined by a 3×1 convolution. These features are concatenated with the corresponding map from the encoder (skip connection) before passing through subsequent residual blocks.

Conditioning is implemented via Feature-wise Linear Modulation (FiLM). A separate embedding network processes the condition $\mathbf{u}_t$ using a 4-block ResNet-style 1D encoder to produce a 32-dimensional state embedding. This is concatenated with a 32-dimensional sinusoidal time embedding of t to form a 64-dimensional global conditioning vector. This vector predicts the affine scale and shift parameters for Group Normalization layers throughout the U-Net. The model is trained using the simulation-free Flow Matching objective for 1000 epochs with a batch size of 64.

5.2 Latent Flow Matching (LFM)

The second baseline, Latent Flow Matching (LFM), adopts a two-stage design inspired by Latent Diffusion Models, decoupling the learning of a compressed spatial representation from the learning of temporal dynamics.

Stage 1: Compression. We first train a convolutional autoencoder to compress the state $\mathbf{u}_t$ into a low-dimensional latent representation $\mathbf{z}_t = E_\psi(\mathbf{u}_t) \in \mathbb{R}^{C_z \times L_z}$. The encoder E_ψ is a hierarchical residual network that reduces the spatial dimension by a factor of 16 (from $N = 256$ to $L_z = 16$) while projecting to $C_z = 64$ latent channels. It utilizes channel multipliers $[1, 2, 4, 8]$ on a base width of 64 and includes self-attention mechanisms at the lowest resolution. The decoder D_ψ mirrors this structure to reconstruct the signal. The autoencoder is trained with an ℓ_2 reconstruction loss and a latent regularization penalty for 100 epochs, after which the weights ψ are frozen.

Stage 2: Latent Dynamics. In the second stage, Flow Matching is applied solely in the latent space. We employ a conditional U-Net backbone identical to the FM baseline in terms of depth and conditioning mechanisms, but adapted to operate on the latent tensor dimensions ($C_z = 64$). The flow matches the trajectory of the encoded latents $\mathbf{z}_{t+\Delta t} = E_\psi(\mathbf{u}_{t+\Delta t})$.

5.3 Parameter Flow Matching (PFM)

Our proposed method, Parameter Flow Matching (PFM), introduces a hybrid approach that combines a deterministic feature extractor with a probabilistic generative model over the network weights. This architecture consists of a fixed deterministic backbone and a flow-based weight generator.

Deterministic Backbone. The feature extractor Φ_θ is a standard 1D U-Net that maps the input state $\mathbf{u}_t$ to a high-dimensional feature basis $\Phi_\theta(\mathbf{u}_t) \in \mathbb{R}^{64 \times N}$. This network utilizes a base width of 32 and channel multipliers $[1, 2, 4, 8]$. The encoder uses strided convolutions (kernel size 4, stride 2)

for downsampling, while the decoder uses nearest-neighbor upsampling. A bottleneck stage with self-attention captures long-range spatial correlations. Unlike the generative baselines, this backbone is deterministic; for each training pair, we solve for the optimal last-layer weights $\mathbf{w}^* \in \mathbb{R}^{64}$ via ridge-regularized linear regression, creating a meta-dataset of weight vectors $\mathcal{D} = \{(\mathbf{u}_t^{(i)}, \mathbf{w}^{*(i)})\}$.

Flow on Weights. The generative component $v_\phi(\mathbf{z}_t, t, \mathbf{u}_t)$ models the distribution of these optimal weights $\mathbf{w}^*$. This flow is parameterized by a Conditional Multi-Layer Perceptron (CondMLP).

- **Architecture:** The MLP consists of four residual blocks with a hidden width of 64, Layer Normalization, SiLU nonlinearity, and dropout.
- **Conditioning:** Similar to the baselines, the state $\mathbf{u}_t$ and time t are embedded into a 64-dimensional global vector. This vector modulates the MLP activations via FiLM layers, allowing the network to predict the velocity of the weight vector distribution.

Training is performed using the simulation-free FM loss on the weights. During inference, we integrate the flow for $T = 20$ ODE steps to sample a weight vector $\hat{\mathbf{w}}$, which is combined with the deterministic basis $\Phi_\theta(\mathbf{u}_t)$ to produce the final prediction.

6 Experiments

6.1 Experimental Setup

Dataset. We generate data using a pseudo-spectral solver with 256 spatial grid points ($N = 256$). The domain length L is randomized $L \sim \mathcal{U}[0.9L_{\text{nom}}, 1.1L_{\text{nom}}]$ to encourage scale invariance. We generate 100 trajectories, utilizing a 80/10/10 split for training, validation, and testing. The inputs and targets are normalized to zero mean and unit variance.

Baselines. We compare PFM against three distinct architectures:

- **Deterministic UNet:** A standard 1D UNet trained with MSE loss. This serves as the upper bound for deterministic accuracy.
- **Pixel-Space Flow Matching (FM):** A conditional FM model trained directly on the state space $\mathbb{R}^N$.
- **Latent Flow Matching (LFM):** A Latent Diffusion-style model that first trains an autoencoder to compress the state, then applies FM in the latent space.

Evaluation Metrics. Let $\mathcal{D}_{\text{test}}$ denote the test set with size M, and let $\{\hat{\mathbf{u}}_i^{(k)}\}_{k=1}^K$ represent the ensemble of K predictions for the ground truth $\mathbf{u}_i$. We define the ensemble mean as $\bar{\mathbf{u}}_i = \frac{1}{K}\sum_{k=1}^K \hat{\mathbf{u}}_i^{(k)}$. We assess performance using:

1. **Mean Squared Error (MSE):** The error of the ensemble mean against the ground truth:

$$\text{MSE} = \frac{1}{M}\sum_{i=1}^{M} \|\mathbf{u}_i - \bar{\mathbf{u}}_i\|_2^2. \tag{9}$$

2. **Spread:** The mean variance of the predictive ensemble, representing aleatoric uncertainty:

$$\text{Spread} = \frac{1}{M}\sum_{i=1}^{M}\left(\frac{1}{K-1}\sum_{k=1}^{K}\|\hat{\mathbf{u}}_i^{(k)} - \bar{\mathbf{u}}_i\|_2^2\right). \tag{10}$$

3. **Spread-Skill Ratio (SSR):** The ratio of the ensemble spread (standard deviation) to the Root Mean Squared Error (RMSE). An SSR of 1.0 indicates perfect calibration:

$$\text{SSR} = \frac{\sqrt{\text{Spread}}}{\sqrt{\text{MSE}}}. \tag{11}$$

6.2 Results

Table 1 presents the quantitative comparison. The deterministic UNet achieves the lowest MSE but provides no uncertainty quantification. The Pixel-Space FM and Latent FM models exhibit high error rates, indicating a failure to converge to the correct conditional manifold; their high spread reflects mode collapse or hallucination rather than calibrated uncertainty.

In contrast, PFM achieves an MSE of 0.022, closely approaching the deterministic baseline (0.014). Crucially, PFM maintains a coherent ensemble with an SSR of 0.705. While slightly under-dispersed (indicating moderate overconfidence), the spread is informative and structurally consistent with the physics.

Table 1: Quantitative comparison of baselines and PFM on the KS one-step prediction task. PFM combines near-deterministic accuracy with meaningful uncertainty calibration, whereas pixel and latent space flows struggle with the complex solution topology.

Model	**MSE** $\downarrow$	**Spread**	**SSR** (Target ≈ 1)	**Paradigm**
Deterministic UNet	**0.014**	–	–	Deterministic
Pixel-Space FM	0.854	0.758	0.815	Generative (State)
Latent FM (LFM)	0.764	0.501	0.578	Generative (Latent)
PFM (Ours)	0.022	0.105	0.705	Generative (Parameter)

Qualitative Analysis. Figure 1 visualizes the predictions. The Pixel-Space FM and LFM models generate noisy, high-frequency artifacts that violate the smoothness constraints of the KS equation (dominated by the fourth-order derivative term). PFM, by operating on the weights of smooth basis functions, guarantees spatial coherence. The uncertainty in PFM predictions is notably concentrated in regions of high non-linearity, such as wavefront mergers, which corresponds to the true dynamic instability of the system.

7 Discussion and Conclusion

The failure of pixel-space and latent-space flow matching in this context highlights the difficulty of learning transport maps in high-dimensional functional spaces where the data manifold is thin and complex. By lifting the generative task to the parameter space of a specialized regressor, PFM simplifies the topology of the target distribution. The weights $\mathbf{w}$ effectively parameterize the local tangent space of the solution operator.

We presented Parameter Flow Matching, a theoretically motivated framework for probabilistic physics emulation. Our results on the Kuramoto–Sivashinsky equation demonstrate that PFM yields high-fidelity predictions with calibrated uncertainty, offering a viable path toward robust uncertainty quantification in scientific machine learning. Future work will extend this approach to autoregressive rollouts and higher-dimensional systems like fluid turbulence.

Administrative Details

Team Contributions.

- **Sungwon Park:** Lead implementation of the PFM architecture, derivation of the meta-learning objective, experimental design, and analysis.
- **Irene Busah:** Implementation of the Standard Flow Matching baseline and KS data generation pipeline.
- **Yuri Xu:** Implementation of the Latent Flow Matching baseline and autoencoder architecture.
- **Huiting Chen:** Literature review, formalization of evaluation metrics, and failure analysis of baselines.

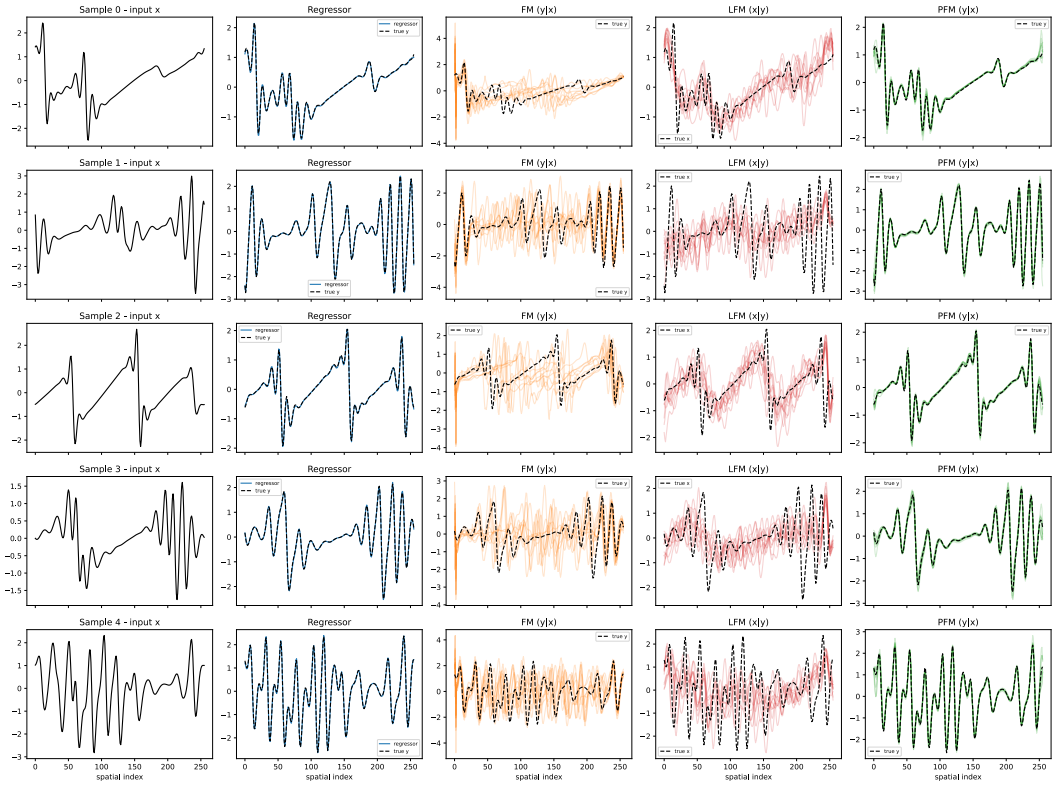


Figure 1: Qualitative comparison of ensemble predictions. The top row shows the ground truth evolution. The deterministic UNet captures the mean mode but lacks probabilistic information. Standard Flow Matching and Latent FM introduce significant noise and unphysical artifacts. PFM (Ours) produces smooth, physically consistent realizations where the ensemble spread (shaded region) accurately reflects the difficulty of the prediction task in unstable regions.

Our implementation code can be found in this repository GitHub.

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